

Projet Data Régression linéaire

Costa Laura – M2 ATDM STAPS

Dataset : tracking gym members

Prédire les calories brûlées

- Construire le meilleur modèle prédictif pour Calories_Burned en utilisant les informations démographiques, physiologiques et liées à l'entraînement
- Dataset : gym_members_exercise_tracking
 - 1 ligne = 1 utilisateur
 - Suppression des doublons et des lignes avec valeurs manquantes
 - **973 lignes, 15 colonnes**

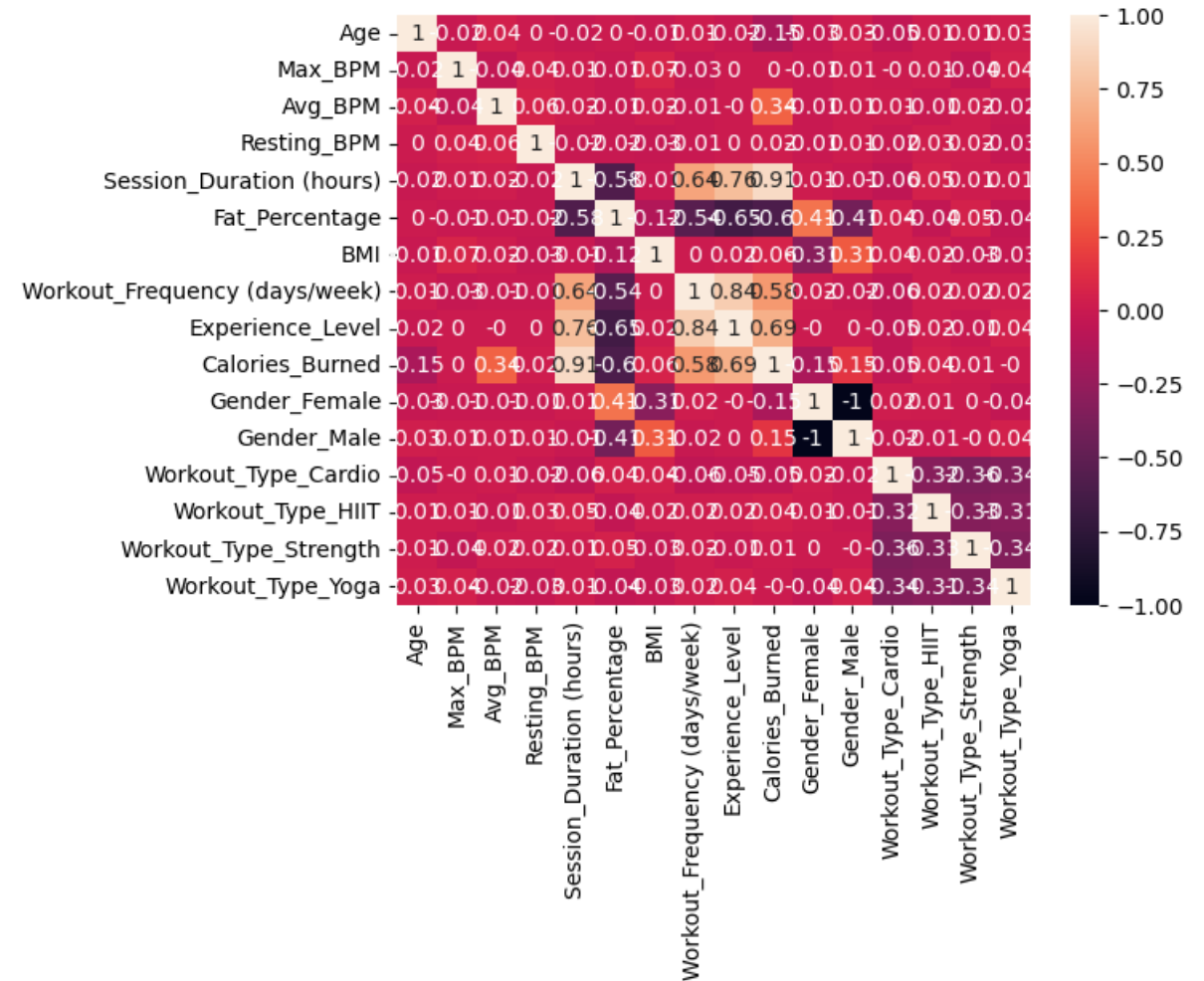
Dataset : tracking gym members

	Age	Weight (kg)	Height (m)	Max_BPM	Avg_BPM	Resting_BPM	Session_Duration (hours)	Calories_Burned	Fat_Percentage	Water_Intake (liters)	BMI
mean	38.683453	73.854676	1.72258	179.883864	143.766701	62.223022	1.256423	905.422405	24.976773	2.626619	24.912127
std	12.180928	21.207500	0.12772	11.525686	14.345101	7.327060	0.343033	272.641516	6.259419	0.600172	6.660879
min	18.000000	40.000000	1.50000	160.000000	120.000000	50.000000	0.500000	303.000000	10.000000	1.500000	12.320000
max	59.000000	129.900000	2.00000	199.000000	169.000000	74.000000	2.000000	1783.000000	35.000000	3.700000	49.840000

Matrice de corrélation pour identifier des features

Correlations remarquables

- Avg_BPM vs Calories_Burned: 0.34
- Session_Duration (hours) vs Calories_Burned: 0.91
- Fat_Percentage vs Calories_Burned: -0.60
- Workout_Frequency (days/week) vs Calories_Burned : 0.58
- Experience_Level vs Calories_Burned: 0.69



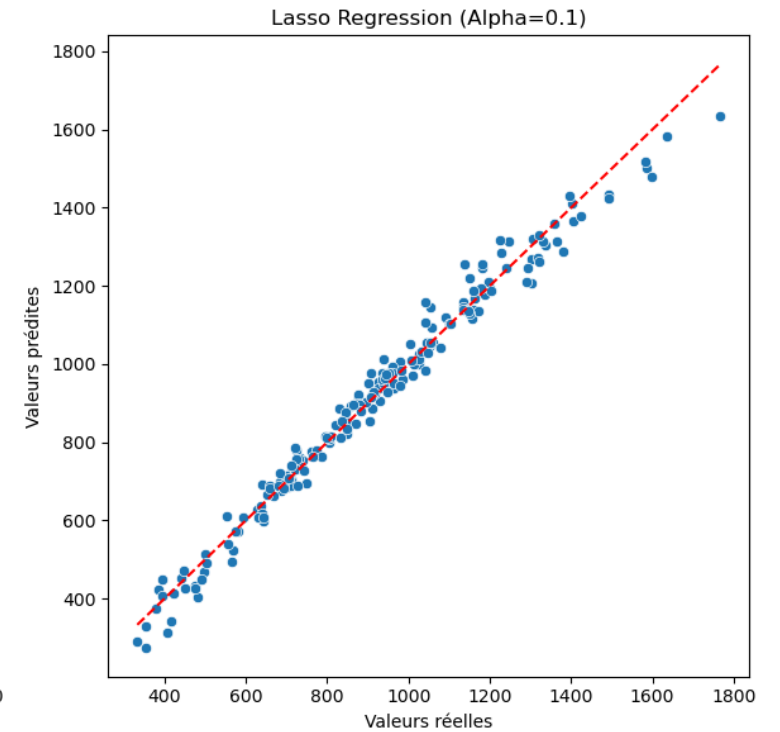
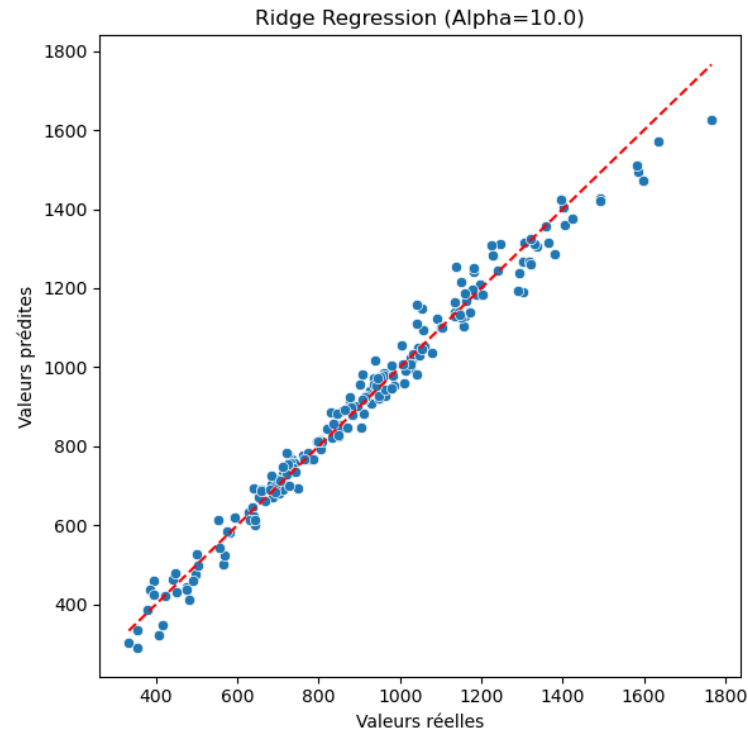
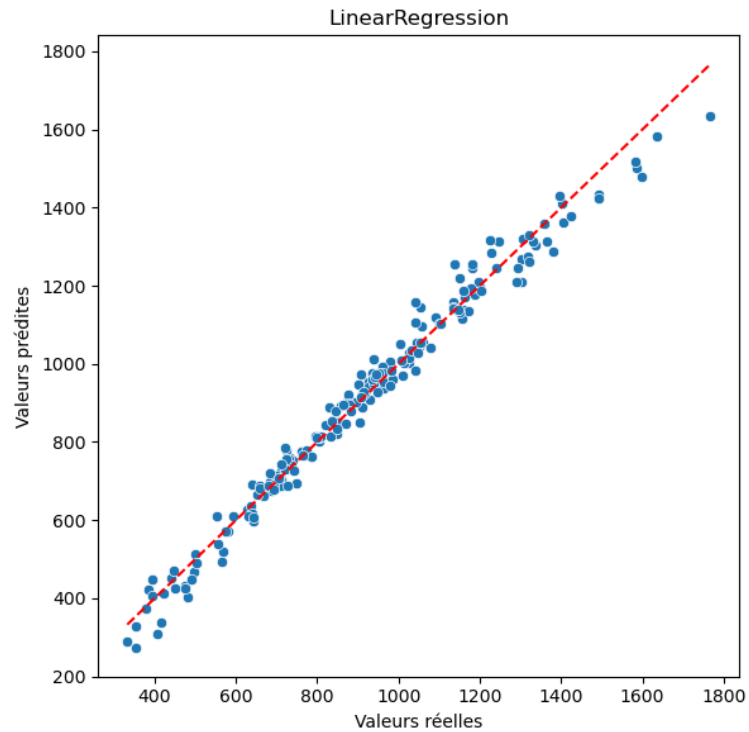
Comparaison de 3 modèles de régression linéaire

- $y = \text{Calories_burned}$
- $X = \{\text{Session_durations(hours)}, \text{Fat_percentage}, \text{Experience_Level}, \text{AVG_BPM}, \text{Age}, \text{Gender}\}$
- Répartition 80%/20%

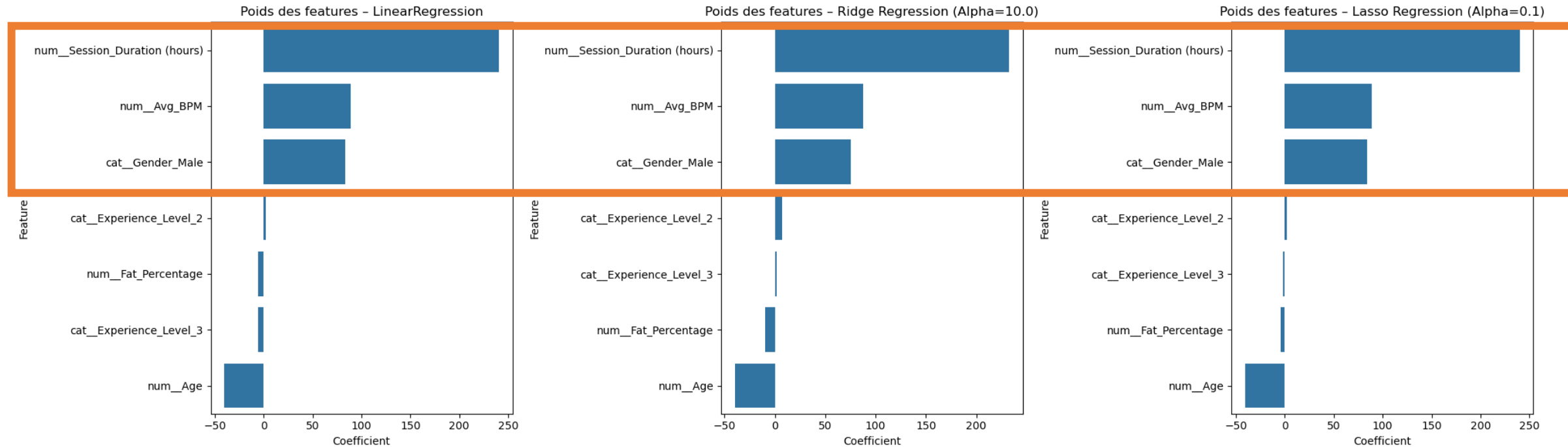
Modèle	R2	RMSE	Intercept
LinearRegression	0.981002	39.810594	859.777496
Ridge Regression (Alpha=10.0)	0.980075	40.770794	860.381804
Lasso Regression (Alpha=0.1)	0.980994	39.819554	858.540667

- 98% de la variance expliquée
- En moyenne, les prédictions se trompent de +/- 40 calories
- Pour un individu moyen, 860 calories

Comparaison des 3 modèles de régression linéaire



Importance de chaque features



Avec Dataiku (avec 12 variables prédictives)

Model Evaluation						> evaluation	> evaluationDataset	> model	> Performance metrics	
Name						date	dataset-name	name	RMSE	R2 Score
☐			Ordinary Least Squares (s2)	OPEN		2025-11-19T13:40:47.40...	gym_members_exercise_tr...	Ordinary Least Squares (s2)	37.064	0.978
☐			Ridge (L2) regression (s2)	OPEN		2025-11-19T13:40:34.05...	gym_members_exercise_tr...	Ridge (L2) regression (s2)	37.070	0.978
☐			Lasso (L1) regression (s2)	OPEN		2025-11-19T13:40:32.64...	gym_members_exercise_tr...	Lasso (L1) regression (s2)	36.881	0.978

- 1 Session_Duration (hours)
- 2 Avg_BPM
- 3 Gender
- 4 Age
- 5 Fat_Percentage
- 6 Workout_Type
- 7 Resting_BPM
- 8 Experience_Level
- 9 Water_Intake (liters)
- 10 Workout_Frequency (days/week)
- 11 Max_BPM
- 12 BMI

